

Interleukin-1 in Human Skin: Dysregulation in Psoriasis

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Cytokine dysregulation is an attractive concept to explain many of the observed abnormalities in psoriasis. IL-1, in particular, can potentiate immune cellular activation, activate fibroblasts, and increase endothelial cell adhesiveness to leukocytes. Here, we review IL-1 regulation in normal and psoriatic skin in vivo in relation to normal skin and cultured keratinocytes. Contrary to expectations, IL-1 functional activity in psoriatic lesions is reduced, not increased, relative to normal skin. The reduction is attributable to the presence of IL-1 inhibitors, reduced IL-1 α levels, and an IL-1 β that lacked function in T-cell assays. IL-1 β protein is actually

significantly increased in psoriatic lesions, but the mechanism of its non-functionality remains unclear. Unlike cultured keratinocytes, which accumulate large, inactive IL-1 β precursors, both normal and psoriatic skin process IL-1 β to a mature form. Novel mechanisms of post-translational processing by epidermis in vivo may generate a novel form of IL-1 β with unknown functions. The marked abnormalities of IL-1 regulation in psoriatic skin suggest that this molecule may be important in normal skin homeostasis. *J Invest Dermatol* 95:24S-26S, 1990

The pathogenesis of psoriasis remains elusive; however, recent hypotheses invoke immunologic cells and cytokines as agents that can explain the multiple features exhibited within psoriatic lesions. The demonstration of a variety of activated immune cells in lesions of psoriasis, the demonstration of cyclosporine A as an effective agent for the treatment of psoriasis, and the demonstration of cytokine-keratinocyte and cytokine-endothelial cell interactions have strengthened such hypotheses.

In addition to immunologic cells releasing inflammatory factors that may propagate psoriasis, keratinocytes produce a number of cytokines, either constitutively or following stimulation, with pro-inflammatory and growth-promoting activities. These include interleukin-1 (IL-1), interleukin-6, interleukin-8, and granulocyte and macrophage colony stimulating factor (GM-CSF). Indeed, we have observed that cytokines released by epidermal cells freshly harvested from untreated lesions of psoriasis can potentiate human T-cell activation in response to allogeneic antigen-presenting cells or can synergize with IL-1 in the stimulation of PHA-stimulated murine thymocytes [1]. Because lesional T-cell activation, either by epidermal non-Langerhans cell macrophages or dermal antigen-presenting cells, is a sensitive indicator of lesional activity, production of cytokines by constitutive elements of the skin could be an important regulator of lesional activity. IL-1 is well known for its ability to potentiate antigen-presenting cell-T-cell interactions with resultant lymphokine release. IL-1 can induce chemotactic cytokine production as well as directly induce adherence of leuko-

cytes to cultured human vascular endothelial cells, and thus may promote entry of leukocytes into psoriatic dermis in vivo. Under certain conditions, exogenous IL-1 may result in keratinocyte proliferation and fibroblast proliferation, both features of psoriatic skin. Because IL-1 is such an excellent candidate for a mediator with pleiotropic effects consistent with the pathophysiologic events observed in psoriasis, increased IL-1 activity has been postulated as a central event in psoriasis. We review here our studies on the regulation of IL-1 in psoriatic skin, in relation to uninvolved and normal skin, as well as in cultured keratinocytes.

Reduced IL-1 Bioactivity in Lesional Skin of Patients with Psoriasis To determine whether IL-1 is elevated in psoriasis, we utilized a standard bioassay where murine thymocytes are stimulated with phytohemagglutinin (PHA) but require IL-1 as a second signal for proliferation. Keratomes were obtained from normal skin, as well as both uninvolved and lesional skin of patients with psoriasis. The keratomes were immediately snap frozen and the proteins extracted in the presence of protease inhibitors. The IL-1 activity, relative to the total protein in each preparation, was assessed by adding the protein extract as the source of IL-1 to the PHA-stimulated thymocytes. To our surprise, relative to normal skin or uninvolved skin of patients with psoriasis, lesional psoriatic skin demonstrated profoundly reduced levels of IL-1 activity [1]. This reduced activity was seen independent of the concentration of cytosolic extract added to the thymocyte stimulation assay. Relative to the uninvolved cytosolic extract, the involved cytosolic extract demonstrated no activity over several logs of protein concentration. We verified the reduced IL-1 bioactivity utilizing another bioassay in which the LBRM.33 cell line responds to IL-1 with increases in the amount of IL-2 produced. Again, cytosolic extracts from lesional skin of psoriasis patients contained remarkably reduced activity relative to normal skin keratome extracts [1,2].

Although IL-1 α Immunoreactive Protein Levels Are Decreased, IL-1 β Immunoreactive Protein Levels Are Increased in Lesional Psoriatic Skin In the course of our studies, we have identified the presence of inhibitors of IL-1 activity that are present

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Abbreviations:

- ELISA: enzyme-linked immunosorbent assays
- FPLC: fast protein liquid chromatography
- GM-CSF: granulocyte and macrophage colony-stimulating factor
- IL-1: interleukin-1
- PHA: phytohemagglutinin

in lesional skin of patients with psoriasis [1-3]. It was, therefore, a distinct possibility that the reduced IL-1 bioactivity was entirely or in part a result of the presence of such inhibitors. We therefore utilized enzyme-linked immunosorbent assays (ELISA) specific for various forms of IL-1 to determine whether the amounts of protein produced in the skin were actually decreased.

At least two distinct forms of IL-1 exist. IL-1 α has been reported to be the predominant product of resting human keratinocytes, whereas IL-1 β is produced by monocytes upon activation [4-6]. Both IL-1 α and IL-1 β are translated as 30-40-kDa polypeptides processed into 17-18kDa polypeptides as they are released from the cell [7-9]. Whereas the IL-1 α precursor has biologic activity [6,10], the precursor form of IL-1 β is biologically inactive [9]. We thus utilized ELISA procedures specific for both IL-1 α and IL-1 β to determine the relative levels of each protein.

In accordance with our bioassay activity data, lesional psoriatic cytosolic extract of keratomes contained twelvefold less IL-1 α per milligram protein relative to cytosolic extracts of normal skin or uninvolved psoriatic skin [11]. However, when IL-1 β immunoreactive protein levels were quantified, we found that the cytosolic extracts of the keratomes from lesional skin contained significantly (twofold) increased amounts of IL-1 β per milligram protein relative to normal skin. Uninvolved skin of psoriatics demonstrated an intermediate level of IL-1 β . Whereas in normal skin the ratio of IL-1 α to IL-1 β is approximately 40:1, in psoriatic lesions the ratio of IL-1 α to IL-1 β became approximately 1:1 (Fig 1).

Despite the Presence of IL-1 β , All IL-1 Activity in Normal and Psoriatic Skin is Due to IL-1 α It was thus not possible to reconcile the almost total absence of IL-1 activity in the bioassay with the increased immunoreactive IL-1 β protein levels. We therefore turned to neutralizing antibodies specific for IL-1 α and IL-1 β to help resolve our findings. We prepared cytosolic extracts of uninvolved and involved keratomes from patients with psoriasis. These were added to murine thymocytes co-stimulated with PHA. In this experiment the decreased level of thymocyte stimulatory activity present in the psoriatic-involved cytosol is apparent, but detectable activity over background was still present (Fig 2). Blockade of IL-1 activity in the extracts with neutralizing antibody to IL-1 α reduced the thymocyte stimulatory activity of both the lesional and non-lesional psoriatic skin completely to background. The specificity of the antibodies is demonstrated in the lower panel where the anti-IL-1 α antibody has no activity against monocyte IL-1 β but has potent activity against recombinant IL-1 α (Fig 2). Thus, all of the IL-1 activity in both the lesional and non-lesional skin could be attributed to IL-1 α , implying that the elevated IL-1 β detected by the ELISA was playing no role in these bioassays. These findings were verified using the LBRM.33 cell line bioassay in which antibodies to IL-1 α completely neutralized bioactivity, whereas antibodies specific for IL-1 β had absolutely no effect [2,12].

Lesional Skin of Patients with Psoriasis Contains Increased IL-1 β mRNA Transcripts, Confirming Up-Regulated IL-1 β Protein The complete neutralization of IL-1 bioactivity by anti-IL-1 α antibodies and the failure of anti-IL-1 β antibodies to neutralize bioactivity despite immunoreactivity by ELISA raised the possibility that we were not detecting IL-1 β in the ELISA. We therefore examined whether messenger RNA levels for IL-1 β were coordinately upregulated. RNA was extracted from keratome biopsies and equal quantities of total RNA were subjected to electrophoresis and Northern analysis. Hybridization of an IL-1 β cDNA probe to the N-terminal precursor region of pro-IL-1 β mRNA demonstrated a 25-fold increase in IL-1 β mRNA in involved psoriatic skin relative to normal skin [1]. This provided strong evidence that the IL-1 β immunoreactive protein present in lesional psoriasis skin was a consequence of increased IL-1 β mRNA transcripts. There is no evidence that these transcripts represented products of a novel gene for IL-1 β , because no differences in migration on the gel were detectable when compared to IL-1 β transcript size from the monocyte cell line, HL-60.

As reported by others [13], IL-1 α mRNA transcripts were not

detectable by Northern analysis in either normal or involved skin of patients with psoriasis [14]. This was in contrast to cultured normal keratinocytes, which demonstrated high levels of both IL-1 α and IL-1 β mRNA transcripts. High IL-1 mRNA corresponded with high levels of both IL-1 α and IL-1 β protein in extracts of cultured keratinocytes, in a ratio of approximately 1:1 (Fig 1).

Because IL-1 α and IL-1 β are coded by different genes on different chromosomes, it may not be unexpected that their mRNA can be regulated differentially under certain conditions. Understanding their relative response elements will be important in the design of drugs designed to interfere with IL-1.

Is IL-1 β Inactive in Human Skin Because It Exists in the Unprocessed, Inactive Precursor State? IL-1 is also regulated post-translationally by proteases. As described above, IL-1 β must be cleaved from the 30-kDa precursor form to the 15-17-kDa processed form in order to become active. This usually occurs as the molecule is released from the cytoplasm into the surrounding extracellular milieu. To determine whether cutaneous IL-1 β becomes active upon secretion or release, keratomes freshly obtained were placed into culture and IL-1 bioactivity and immunoreactivity analyzed in the supernatant 48 h later. IL-1 β immunoreactive protein, as detected by ELISA, could be observed to be released from the psoriasis keratome, but the amount released by normal keratomes or cultured keratinocytes was below the limit of detection. Both the normal keratome and the psoriatic keratome demonstrated IL-1 bioactivity as measured by LBRM.33 cells. However, all of the released activity from the psoriatic involved keratome was neutralized only with anti-IL-1 α antibodies and not with anti-IL-1 β antibodies or normal rabbit serum. This implied that IL-1 β released by psoriatic involved skin remained dysfunctional [1].

Because the IL-1 β remained inactive upon release from the cells, it was still not clear whether the IL-1 β was present in a processed or precursor state. We therefore fractionated cytosolic proteins from keratomes of the lesions in patients with psoriasis by molecular weight. Fast protein liquid chromatography (FPLC) was employed with a TSK G2000SW column. IL-1 β immunoreactivity by ELISA was then tested in the various fractions, revealing that the majority of the IL-1 β in the cytosol was in a processed and not the large precursor state. IL-1 α by ELISA was also present in a processed state. The reduced activity of the IL-1 β could not be attributed to the presence of the IL-1 inhibitor because this column specifically separated the IL-1 inhibitory activity from the IL-1 β immunoreactive protein. Under these conditions, fractions containing IL-1 β immunoreactivity were distinct from IL-1 inhibitor-containing fractions and still lacked IL-1 β neutralizable bioactivity [12].

Figure 1 summarizes our findings. Whereas normal skin contains

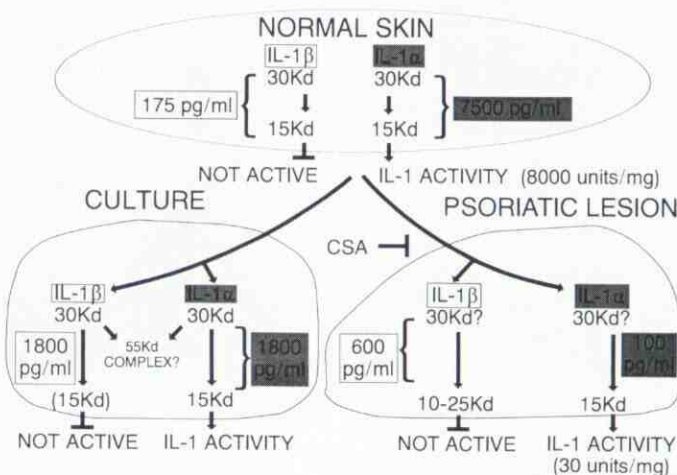


Figure 1. Model for IL-1 regulation in human skin and cultured keratinocytes.

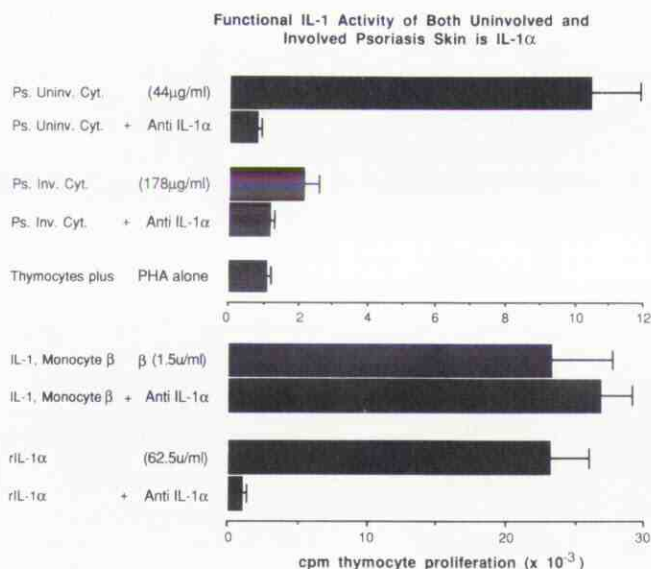


Figure 2. IL-1 activity in the thymocyte proliferation co-stimulation assay of both uninvolved and involved psoriatic skin abrogated with neutralizing antibody to IL-1 α . Fifty microliters of uninvolved cytosol (Ps. Uninv. Cyt.) at a protein concentration of 44 µg/ml and psoriatic involved cytosol (Ps. Inv. Cyt.) were added in the thymocyte proliferation assay the presence or absence of neutralizing antibody to anti-IL-1 α . The specificity of IL-1 α antibodies was demonstrated to have minimal activity against monocyte-derived IL-1 β but complete inhibitory activity against recombinant human IL-1 α .

both IL-1 β and IL-1 α , IL-1 α is the predominant species at a ratio of approximately 40:1 IL-1 α to IL-1 β . Both are present in the precursor and processed forms, yet IL-1 β is inactive in T-cell assays, whereas IL-1 α comprises the entire bioactive IL-1. If normal-appearing skin develops psoriasis, however, IL-1 α is downregulated, whereas IL-1 β is upregulated, resulting in a ratio of approximately 1:1. Despite the elevation in IL-1 β , psoriatic lesional IL-1 β is also inactive, and IL-1 α , although reduced, comprises all the bioactive material. The IL-1 β in the psoriatic lesion may be of a novel form, because it exists as the processed molecule without having attained bioactivity. If normal adult skin is placed into culture, cultured keratinocytes do not reflect the status of either normal skin or the psoriatic lesion. In this situation, both the IL-1 α and IL-1 β are present in relatively high amounts, and mRNA transcripts for both are also upregulated relative to normal skin. In contrast to either normal skin or psoriatic lesions, the IL-1 β in cultured keratinocytes appears to exist predominantly as the 30-kDa precursor, as well as a larger, potentially complexed form of approximately 55 kDa. Again, all activity in cultured keratinocytes is IL-1 α .

Thus, our data do not support the hypothesis that multiple pathophysiologic events potentially attributable to IL-1 in psoriasis are related to elevated IL-1 activity. However, the profound dysregula-

tion of IL-1 that occurs in psoriatic lesions suggests that understanding the regulatory mechanisms governing IL-1 production and processing will provide a better understanding of cytokines and their relationship to immunologic cells and constitutive elements of the skin in this disease.

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